

Automatic Multi-Speed Fan Controller (Auto-Mode Fan)

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1. Executive Summary & Objective

This technical report provides a detailed analysis, mathematical modeling, and Proteus simulation review for the Automatic Multi-Speed Fan Controller (Auto-Mode Fan). The system monitors ambient temperature in real time, amplifies the low-level transducer signal, quantizes the thermal levels via a 4-level Flash ADC comparator array, synthesizes digital selector lines using combinational logic, and switches multi-tier drive voltages using an analog multiplexer (CD4052) paired with a BJT current booster and flyback protection network.

2. Circuit Architecture & Block Diagram

- Stage 1: Transduction & Amplification:** LM35 linear sensor (10 mV/°C) conditioned by an LM741 Non-Inverting Op-Amp ($A_v = 7.8$).
- Stage 2: 4-Level Flash ADC Conversion:** Resistor ladder and 4 parallel LM741 comparators generating thermometer code outputs (q_0, q_1, q_2, q_3).
- Stage 3: Combinational Selector Synthesis:** Logic network (OR, AND, NOT) synthesizing multiplexer address lines S_0 and S_1 .
- Stage 4: Multiplexed Speed Control & Power Drive:** CD4052 4-to-1 MUX selecting preset voltage taps, amplified by an NPN BJT driver with flyback diode protection.

3. Mathematical Modeling & Stage-by-Stage Analysis

Stage 1: Temperature Transduction & Op-Amp Amplification

The LM35 produces an analog voltage that scales linearly with temperature at 10 mV/°C:

$$V_{\text{sensor}}(T) = 10 \text{ mV/}^{\circ}\text{C} \times T$$

To condition this millivolt-level output for precise comparison, an LM741 operational amplifier is configured in a non-inverting topology with feedback resistor $R_{10} = 68 \text{ k}\Omega$ and ground resistor $R_9 = 10 \text{ k}\Omega$. The closed-loop gain (A_v) and amplified voltage $V_{\text{amp}}(T)$ are calculated as:

$$A_v = 1 + (R_{10} / R_9) = 1 + (68 \text{ k}\Omega / 10 \text{ k}\Omega) = 1 + 6.8 = 7.8$$
$$V_{\text{amp}}(T) = A_v \times V_{\text{sensor}}(T) = 7.8 \times (0.010 \times T) = 0.078 \times T \text{ [V]}$$

• Operating Verification at $T = 25^{\circ}\text{C}$:

$$V_{\text{sensor}} = 25 \times 10 \text{ mV} = 0.25 \text{ V} \rightarrow V_{\text{amp}} = 7.8 \times 0.25 \text{ V} = 1.95 \text{ V (Output Node U10(OP))}$$

Stage 2: 4-Level Flash ADC Conversion

The amplified voltage V_{amp} is fed into the non-inverting (+) inputs of 4 parallel LM741 comparators (U6, U1, U2, U3). A resistor ladder consisting of four matched resistors ($R_1, R_2, R_3, R_4 = 10 \text{ k}\Omega$) divides the reference voltage into 4 equal thresholds ($V_{\text{ref}} \approx 0.8 \text{ V step}$):

- Each comparator evaluates: $q_n = \text{HIGH (1)}$ when $V_{\text{amp}} > V_{\text{ref},n}$; otherwise LOW (0).

Stage 3: Combinational Logic for Multiplexer Addressing

Instead of a bulky priority encoder IC, a compact discrete gate network synthesizes the 2-bit channel select lines (S_0, S_1) for the analog multiplexer:

$$S_0 = q_3 + (q_1 \cdot q_2') \quad [\text{Synthesized via U7 NOT, U8 AND, U5 OR}]$$
$$S_1 = q_2 + q_3 \quad [\text{Synthesized via U4 OR}]$$

Stage 4: Multiplexed Speed Control & Power Driving

A CD4052 analog multiplexer selects among 4 discrete operating voltages generated by a 4-resistor ladder (R5, R6, R7, R8 = 100 Ω each):

- Selection 00 (X0): Lowest speed / Standby tap.
- Selection 01 (X1): Low-intermediate speed tap.
- Selection 10 (X2): High-intermediate speed tap.
- Selection 11 (X3): Maximum speed / Full VCC tap.

BJT Power Booster & Protection: Because multiplexer output current is limited and unable to drive motor torque directly, an NPN transistor (Q1) is configured as a Common Emitter / Emitter Follower current amplifier. A flyback protection diode (D1) is placed antiparallel across the motor to dissipate inductive back-EMF spikes when switching off, protecting Q1 from overvoltage breakdown.

4. Digital Truth Table & Speed Mode Matrix

Thermal Level	Comparator State (q3 q2 q1 q0)	S1 (MSB)	S0 (LSB)	Selected MUX Ch	Fan Speed & Operating Mode
T < Level 1	0 0 0 0	0	0	X0 (Tap 0)	Speed 0 / Idle (Minimum V)
Level 1 ≤ T < Level 2	0 0 1 1 (or 0 0 0 1)	0	1	X1 (Tap 1)	Speed 1 / Low Speed
Level 2 ≤ T < Level 3	0 1 1 1	1	0	X2 (Tap 2)	Speed 2 / Medium Speed
T ≥ Level 3 (Max)	1 1 1 1	1	1	X3 (Tap 3)	Speed 3 / Maximum Cooling

5. Bill of Materials (BOM) & Component Parameters

Designator	Component Name	Parameter / Value	Functional Role in Circuit
U11	LM35 Temperature Sensor	10 mV/°C Precision	Transduces ambient temperature to linear mV voltage
U10	LM741 Operational Amplifier	Av = 7.8 (Non-Inverting)	Amplifies sensor voltage to drive Flash ADC
R9, R10	Metal Film Resistors	10 kΩ, 68 kΩ	Feedback network setting op-amp voltage gain
R1 - R4	ADC Resistor Ladder	10 kΩ matched (4x)	Sets reference threshold voltages for comparators
U6, U1, U2, U3	LM741 Comparators	4x Parallel Comparators	Quantizes analog signal into thermometer digital code
U7, U8, U4, U5	Logic Gates	NOT, AND, OR, OR	Synthesizes MUX select lines: S0 = q3 + q1·q2', S1 = q2 + q3
U12	CD4052 CMOS Multiplexer	Dual 4-Channel Analog MUX	Selects analog voltage tap according to S1, S0
R5 - R8	Speed Divider Ladder	100 Ω matched (4x)	Generates 4 discrete voltage steps for fan

			speeds
Q1	NPN BJT Transistor	Power Booster	Provides high current gain to drive fan motor torque
D1	Flyback Rectifier Diode	1N4007 / Fast Recovery	Suppresses inductive kickback spikes from motor
MOTOR	DC Fan Motor	Inductive Load	Provides variable airflow based on selected voltage tap

6. Engineering Observations & Proteus Simulation Review

- **Gain Optimization:** The revised gain of $A_v = 7.8$ ($R_{10}=68k\Omega$, $R_9=10k\Omega$) matches the 4-level Flash ADC input span efficiently, providing 1.95 V output at 25°C without saturating the op-amp rails.
- **Hardware Efficiency:** Replacing a full priority encoder IC with 4 discrete logic gates reduced circuit footprint while achieving accurate binary addressing (S0, S1).
- **Torque & Power Solved:** The addition of the BJT amplifier resolved the initial weak-torque problem under MUX drive, while the flyback diode safely prevented inductive breakdown across the transistor.